

Product Geometry for Predictable Linear Reinforcement

Elephant Walks, Universality and Critical Windows

Research draft

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Abstract

We study discrete-time processes whose conditional drift has the predictable linear reinforcement form

$$\mathbb{E}[\xi_{n+1} \mid \mathcal{F}_n] = a_n S_n / n, \quad S_n = \sum_{k \leq n} \xi_k,$$

with bounded predictable environment a_n . The point is not a new variant of one random walk, but a classification principle. Once the environment is fixed, the asymptotic behaviour is governed by two canonical quantities: the multiplicative normalisation

$$A_n = \prod_{k < n} \left(1 + \frac{a_k}{k}\right)$$

and the quadratic variation of the martingale S_n/A_n . The main theorem proves a product-bracket classification for predictable linear reinforcement schemes satisfying explicit regularity and Lindeberg conditions. Infinite bracket gives Gaussian martingale limits with deterministic time changes determined by the ratios of the product and bracket; finite bracket gives a terminal martingale law. Predictable elephant random walks are then a canonical application: their bracket decomposes as a product-determined clock minus a negligible self-interaction term. This yields the classical diffusive, critical and superdiffusive regimes as corollaries and exposes logarithmic critical windows in which $a_n \rightarrow 1/2$ produces three different normalising scales. We also prove rigidity: schemes with equivalent product and bracket geometry have the same scaling limits. Counterexamples show why convergence of the coefficient alone cannot classify the asymptotics.

Keywords: predictable reinforcement; elephant random walk; martingales; universality; phase transition; regular variation.

1 Introduction

Many reinforced and self-interacting processes have a linear first-order drift: after a suitable projection, the next increment has conditional mean proportional to the present state divided by time. In the simplest scalar form,

$$\mathbb{E}[\xi_{n+1} \mid \mathcal{F}_n] = a_n \frac{S_n}{n}, \quad S_n = \sum_{k \leq n} \xi_k,$$

where the coefficient a_n is predictable. Fixed a_n gives the classical linear reinforcement exponents. Predictable a_n , however, creates a different problem: small changes in the drift accumulate over logarithmic time, so the limiting coefficient alone cannot determine the asymptotics.

This paper isolates the invariant behind this phenomenon. The conditional drift is removed by the multiplicative product

$$A_n = \prod_{k < n} \left(1 + \frac{a_k}{k}\right),$$

and the remaining fluctuations are governed by the quadratic variation of the martingale S_n/A_n . The product and the bracket are therefore the two canonical coordinates of a predictable linear reinforcement scheme. The main result proves that, under explicit regularity and Lindeberg conditions, these coordinates classify the scaling limits: infinite bracket gives Gaussian limits with deterministic time change, while finite bracket gives a terminal martingale law.

The elephant random walk enters as the basic complete-memory model where this principle can be pushed to a concrete phase diagram. At time $n + 1$, the walker chooses a past increment uniformly and either repeats or reverses it. For a fixed memory parameter the model has diffusive, critical and superdiffusive regimes (Schütz and Trimper, 2004; Coletti et al., 2017; Baur and Bertoin, 2016; Bercu, 2018). If the repeat–reverse coefficient is predictable, the same martingale product appears, but the bracket contains a self-interaction correction. Showing that this correction is negligible in the appropriate regimes turns the abstract classification into explicit ERW limits.

The point is not that p has been replaced by p_n . The point is that the phase boundary is product-geometric. For example,

$$a_n = \frac{1}{2} + \frac{c}{\log n}$$

has the same pointwise limit $1/2$ for every c , but it produces three different bracket geometries and hence three different normalising scales. This logarithmic critical window is the simplest place where the hidden geometry is visible: convergence of the environment is too crude, while the product and bracket see the phase transition.

This perspective separates the present work from several neighbouring theories. Classical ERW methods rely on a fixed exponent. Urn methods rely on a fixed or asymptotically stable replacement scheme (Janson, 2004, 2005; Laruelle and Pagès, 2013). Reinforced processes exhibit many memory mechanisms (Pemantle, 2007; Laulin, 2022), and stochastic approximation studies adaptive parameters through their own limiting recursion (Kushner and Yin, 2003). Here the object being classified is the predictable reinforcement environment itself, through the product–bracket geometry it induces.

The paper is organised as follows. Section 2 defines predictable linear reinforcement schemes and then specialises them to predictable elephant walks. Section 3 states the master classification theorem and the rigidity principle. Section 4 proves the martingale representation and the classification. Section 5 derives the ERW phase diagram. Section 6 gives counterexamples and the logarithmic critical window. Section 7 records learning rules only as examples of predictable environments, and Section 8 collects open problems.

2 Predictable Linear Reinforcement

Let $(\Omega, \mathcal{F}, \mathbb{P})$ carry a filtration $(\mathcal{F}_n)_{n \geq 0}$. A scalar predictable linear reinforcement scheme is an adapted process

$$S_n = S_0 + \sum_{k=1}^n \xi_k$$

with square-integrable increments such that, for a bounded predictable environment $(a_n)_{n \geq 1}$,

$$\mathbb{E}[\xi_{n+1} \mid \mathcal{F}_n] = a_n \frac{S_n}{n}, \quad n \geq 1.$$

Throughout we assume that, for some $\varepsilon \in (0, 1)$,

$$-1 + \varepsilon \leq a_n \leq 1 - \varepsilon.$$

This is the natural non-degeneracy condition for the multiplicative normalisation. Define

$$A_0 = A_1 = 1, \quad A_{n+1} = A_n \left(1 + \frac{a_n}{n}\right),$$

The product A_n is the first invariant of the scheme. The second invariant is the bracket of the martingale $M_n = S_n/A_n$, denoted by

$$V_n = \langle M \rangle_n.$$

The pair (A_n, V_n) is the canonical product–bracket geometry.

All convergence assumptions in the master theorem are annealed statements unless explicitly stated otherwise. In the deterministic, or externally sampled and then conditioned, case the same assumptions may be read quenched, path by path. This distinction matters only for how the product–bracket geometry is verified; the martingale classification itself is identical.

The predictable elephant random walk is the main example. It has increments $X_n \in \{-1, 1\}$, position $S_n = \sum_{j=1}^n X_j$, and $S_0 = 0$. At time $n + 1$, choose K_n conditionally uniform on $\{1, \dots, n\}$. If p_n is $\mathcal{F}_n$ -measurable and $a_n = 2p_n - 1$, then conditional on $\mathcal{F}_n$ and K_n ,

$$X_{n+1} = \begin{cases} X_{K_n}, & \text{with probability } p_n, \\ -X_{K_n}, & \text{with probability } 1 - p_n. \end{cases}$$

Consequently

$$\mathbb{E}[X_{n+1} \mid \mathcal{F}_n] = a_n \frac{S_n}{n},$$

so the ERW is a predictable linear reinforcement scheme. In this case the bracket has the explicit decomposition

$$V_n = Q_n - R_n, \quad Q_n = \sum_{k=1}^{n-1} \frac{1}{A_{k+1}^2}, \quad R_n = \sum_{k=1}^{n-1} \frac{a_k^2 S_k^2}{k^2 A_{k+1}^2}.$$

Thus the ERW phase diagram reduces to proving that R_n is negligible relative to the product-determined clock Q_n .

Definition 1 (Admissible product–bracket geometry). A predictable linear reinforcement scheme is called admissible on $[0, 1]$ if $V_n \rightarrow \infty$ in probability and there exist deterministic functions $r, q : (0, 1] \rightarrow \mathbb{R}$, with q nondecreasing and $r(1) = q(1) = 1$, such that for every $\eta > 0$,

$$\sup_{\eta \leq t \leq 1} \left| \frac{A_{[nt]}}{A_n} - r(t) \right| \xrightarrow{\mathbb{P}} 0, \quad \sup_{\eta \leq t \leq 1} \left| \frac{V_{[nt]}}{V_n} - q(t) \right| \xrightarrow{\mathbb{P}} 0,$$

$$\max_{k \leq n} \frac{|\Delta M_k|}{\sqrt{V_n}} \xrightarrow{\mathbb{P}} 0.$$

It is called tight at the origin if, for every $\delta > 0$,

$$\lim_{\eta \downarrow 0} \limsup_{n \rightarrow \infty} \mathbb{P} \left(\sup_{0 \leq t \leq \eta} \frac{|S_{[nt]}|}{A_n \sqrt{V_n}} > \delta \right) = 0.$$

Remark 1. The definition is deliberately formulated in terms of A_n and V_n , not in terms of a limiting coefficient a . This is the point of the classification: the coefficient path is not itself the invariant. Two paths with the same limit may generate different products, and two paths with different local behaviour may nevertheless lie in the same product–bracket class.

3 The Classification Theorem

Theorem 1 (Master classification). *Let (S_n) be a square-integrable predictable linear reinforcement scheme with bounded environment (a_n) .*

1. *If the scheme is admissible and tight at the origin in the sense of Definition 1, and if the process $G(t) = r(t)B(q(t))$, $t > 0$, $G(0) = 0$, has a continuous version, then the linearly interpolated walk satisfies*

$$\left(\frac{S_{[nt]}}{A_n \sqrt{V_n}} \right)_{0 \leq t \leq 1} \Rightarrow (G(t))_{0 \leq t \leq 1}$$

in $C[0, 1]$, where B is a standard Brownian motion.

2. *If $V_\infty = \lim_n V_n < \infty$ almost surely and $\mathbb{E}[V_\infty] < \infty$, then $M_n = S_n/A_n$ converges almost surely and in L^2 to a finite random variable M_∞ . Consequently*

$$\frac{S_n}{A_n} \longrightarrow M_\infty \quad \text{almost surely and in } L^2.$$

Thus every limit in the paper is obtained by identifying the asymptotic geometry of A_n and V_n . The next result records the rigidity built into this classification.

Theorem 2 (Rigidity of the product–bracket geometry). *Consider two predictable linear reinforcement schemes $e = 1, 2$, with canonical quantities $(A_n^{(e)}, V_n^{(e)})$. Suppose both are admissible and tight at the origin, and*

$$\frac{A_n^{(1)}}{A_n^{(2)}} \xrightarrow{\mathbb{P}} c_A \in (0, \infty), \quad \frac{V_n^{(1)}}{V_n^{(2)}} \xrightarrow{\mathbb{P}} c_V \in (0, \infty),$$

and that their local ratios have the same limits:

$$\frac{A_{[nt]}^{(e)}}{A_n^{(e)}} \xrightarrow{\mathbb{P}} r(t), \quad \frac{V_{[nt]}^{(e)}}{V_n^{(e)}} \xrightarrow{\mathbb{P}} q(t), \quad e = 1, 2.$$

Then both environments have the same functional limit after their own canonical normalisation. More precisely, if

$$G(t) = r(t)B(q(t)), \quad t > 0, \quad G(0) = 0,$$

has a continuous version, then for $e = 1, 2$,

$$\frac{S_{[nt]}^{(e)}}{A_n^{(e)} \sqrt{V_n^{(e)}}} \Rightarrow G(t) \quad \text{in } C[0, 1].$$

In particular, the scaling limit is a function only of the product–bracket geometry (r, q) , not of the particular predictable rule that generated it.

Remark 2. The theorem is not stated as a global if-and-only-if theorem. Without additional regularity, failure of product or bracket equivalence does not by itself specify which alternative limit appears. Section 6 gives explicit families showing that the invariants are indispensable and identifies the controlled classes where converses are valid.

4 Martingale and Bracket Theory

Lemma 1 (Exact martingale normalisation). *The process $M_n = S_n/A_n$ is a martingale with respect to $(\mathcal{F}_n)$, and*

$$\Delta M_{n+1} = \frac{\xi_{n+1} - a_n S_n/n}{A_{n+1}}.$$

Proof. By the defining drift condition,

$$\mathbb{E}[S_{n+1} \mid \mathcal{F}_n] = S_n + a_n \frac{S_n}{n} = \left(1 + \frac{a_n}{n}\right) S_n.$$

Dividing by $A_{n+1} = A_n(1 + a_n/n)$ gives $\mathbb{E}[M_{n+1} \mid \mathcal{F}_n] = M_n$. The increment formula follows by subtracting M_n . $\square$

Lemma 2 (ERW bracket decomposition). *For the predictable elephant random walk, for all $n \geq 1$,*

$$\langle M \rangle_n = \sum_{k=1}^{n-1} \frac{1 - a_k^2 S_k^2/k^2}{A_{k+1}^2} = Q_n - R_n.$$

Proof. By Lemma 1,

$$\mathbb{E}[(\Delta M_{k+1})^2 \mid \mathcal{F}_k] = \frac{\mathbb{E}[(X_{k+1} - a_k S_k/k)^2 \mid \mathcal{F}_k]}{A_{k+1}^2}.$$

Since $X_{k+1}^2 = 1$ and $\mathbb{E}[X_{k+1} \mid \mathcal{F}_k] = a_k S_k/k$, the conditional variance equals $1 - a_k^2 S_k^2/k^2$. Summation gives the formula. $\square$

Lemma 3 (ERW diffusive second moment bound). *For the predictable elephant random walk, if $a_n \leq \bar{a} < 1/2$ for all n , then*

$$\sup_{n \geq 1} \frac{\mathbb{E}[S_n^2]}{n} < \infty.$$

Proof. Let $W_n = \mathbb{E}[S_n^2]$. From the conditional drift,

$$\mathbb{E}[S_{n+1}^2 \mid \mathcal{F}_n] = \left(1 + \frac{2a_n}{n}\right) S_n^2 + 1 \leq \left(1 + \frac{2\bar{a}}{n}\right) S_n^2 + 1.$$

Iteration of this scalar recursion gives

$$W_n \leq C n^{2\bar{a}} \left(1 + \sum_{k < n} k^{-2\bar{a}}\right) \leq C n,$$

because $2\bar{a} < 1$. $\square$

Proof of Theorem 1. Assume first that the environment is admissible and tight at the origin. Fix $\eta > 0$. On $[\eta, 1]$, the martingale

$$N_n(t) = \frac{M_{[nt]} - M_1}{\sqrt{V_n}}$$

has jumps bounded by

$$\max_{k \leq n} \frac{|\Delta M_k|}{\sqrt{V_n}},$$

which tend to zero by admissibility. Its predictable bracket is

$$\langle N_n \rangle(t) = \frac{V_{[nt]}}{V_n} \xrightarrow{\mathbb{P}} q(t) \quad \text{uniformly for } \eta \leq t \leq 1.$$

The martingale functional central limit theorem (Hall and Heyde, 1980, Chapter 3) yields

$$N_n \Rightarrow B(q(t)) \quad \text{on } [\eta, 1].$$

Moreover $A_n M_1 / (A_n \sqrt{V_n}) = M_1 / \sqrt{V_n} \rightarrow 0$, so

$$\frac{S_{[nt]}}{A_n \sqrt{V_n}} = \frac{A_{[nt]}}{A_n} \frac{M_{[nt]}}{\sqrt{V_n}} \Rightarrow r(t)B(q(t)) \quad \text{on } [\eta, 1]$$

by the continuous mapping theorem. The tightness-at-the-origin condition makes the contribution of $t \leq \eta$ uniformly negligible after $\eta \downarrow 0$. Since the asserted limit has a continuous version with value 0 at the origin, convergence holds in $C[0, 1]$ for the linearly interpolated walk.

If $V_\infty < \infty$, then

$$\sup_n \mathbb{E}[M_n^2] = \mathbb{E}[M_1^2] + \sup_n \mathbb{E}[\langle M \rangle_n] \leq \mathbb{E}[M_1^2] + \mathbb{E}[V_\infty] < \infty.$$

The martingale convergence theorem gives $M_n \rightarrow M_\infty$ almost surely and in L^2 . $\square$

Proof of Theorem 2. Apply Theorem 1 to each environment. The hypotheses give the same limiting functions r and q . The constants c_A and c_V change the canonical normalising constants but disappear because each walk is normalised by its own $A_n^{(e)} \sqrt{V_n^{(e)}}$. Hence both limits are the same process $r(t)B(q(t))$. $\square$

5 The Phase Diagram

5.1 Diffusive environments

Definition 2 (Uniform diffusive product regularity). An environment is uniformly diffusive with exponent $a < 1/2$ if $a_n \leq \bar{a} < 1/2$ and, for every $\alpha > 0$, there is a deterministic constant $C_\alpha < \infty$ such that, with probability tending to one,

$$\frac{A_n}{A_k} \leq C_\alpha \left(\frac{n}{k}\right)^{a+\alpha}, \quad \frac{A_k}{A_n} \leq C_\alpha \left(\frac{k}{n}\right)^{a-\alpha} \quad (1 \leq k \leq n).$$

Lemma 4 (Product classification below criticality). *For a deterministic environment, assume $a_n \rightarrow a < 1/2$, $a_n \leq \bar{a} < 1/2$, and $-1 + \varepsilon \leq a_n \leq 1 - \varepsilon$. Then*

$$A_n = n^a \ell_n,$$

where (ℓ_n) is slowly varying:

$$\frac{\ell_{[nt]}}{\ell_n} \rightarrow 1 \quad (t > 0).$$

Consequently

$$Q_n \sim \frac{n}{(1-2a)A_n^2}, \quad A_n \sqrt{Q_n} \sim \sqrt{\frac{n}{1-2a}}.$$

Proof. The identity

$$\log \frac{A_n}{n^a} = \sum_{k < n} \frac{a_k - a}{k} + O(1)$$

has a summable Taylor remainder because a_k is bounded away from -1 . For fixed $t > 0$, the difference between the same expression at $[nt]$ and at n is bounded by

$$\sup_{k \geq n(t \wedge 1)/2} |a_k - a| \sum_{k \gtrsim n} \frac{1}{k} + o(1),$$

which tends to zero. Thus A_n/n^a is slowly varying. Karamata's theorem for regularly varying sequences gives

$$\sum_{k < n} \frac{1}{A_{k+1}^2} \sim \frac{n}{(1-2a)A_n^2}.$$

□

Corollary 1 (Diffusive ERW classification). *For the predictable elephant random walk, assume that $A_n = n^a \ell_n$ with ℓ_n slowly varying, that the environment is uniformly diffusive in the sense of Definition 2, and that $a_n \leq \bar{a} < 1/2$. Then*

$$\frac{S_n}{\sqrt{n}} \Rightarrow \mathcal{N}\left(0, \frac{1}{1-2a}\right).$$

If, in addition, the product ratios satisfy

$$\frac{A_{[nt]}}{A_n} \rightarrow t^a$$

locally uniformly on $(0, 1]$, then the linearly interpolated process satisfies

$$\left(\frac{S_{[nt]}}{\sqrt{n}}\right)_{0 \leq t \leq 1} \Rightarrow \left(t^a B\left(\frac{t^{1-2a}}{1-2a}\right)\right)_{0 \leq t \leq 1}.$$

Proof. Regular variation gives $Q_n \sim n\{(1-2a)A_n^2\}^{-1}$ in probability. Let E_n be the high-probability event where the uniform product bounds hold. On E_n , the self-interaction correction satisfies

$$\mathbb{E}\left[\frac{R_n}{Q_n} \mathbf{1}_{E_n}\right] \leq Cn^{2a+2\alpha-1} \sum_{k < n} \frac{\mathbb{E}[S_k^2]}{k^{2a+2\alpha+2}} \leq Cn^{2a+2\alpha-1} \sum_{k < n} k^{-2a-2\alpha-1} \rightarrow 0$$

after choosing $\alpha > 0$ with $a + \alpha < 1/2$. Since $0 \leq R_n/Q_n \leq 1$, this proves $R_n/Q_n \rightarrow 0$ in probability after adding the probability of the complementary event. The same estimate with upper limit $[nt]$ gives

$$\sup_{\eta \leq t \leq 1} \frac{R_{[nt]}}{Q_n} \xrightarrow{\mathbb{P}} 0 \quad (\eta > 0).$$

By Lemma 2, $V_n = Q_n - R_n$, so $V_n/Q_n \rightarrow 1$ in probability and the same local ratios hold for V . Moreover,

$$\frac{Q_{[nt]}}{Q_n} \rightarrow t^{1-2a}$$

locally uniformly on $(0, 1]$, and the assumed product-ratio convergence gives $r(t) = t^a$. The jump condition follows from

$$\max_{k \leq n} \frac{|\Delta M_k|}{\sqrt{V_n}} \leq 2 \max_{k \leq n} \frac{1}{A_{k+1}\sqrt{V_n}} \xrightarrow{\mathbb{P}} 0.$$

Finally, for $m = [n\eta]$, Lemma 3 and Doob's inequality applied to the martingale M give

$$\mathbb{E}\left[\sup_{0 \leq t \leq \eta} \left(\frac{S_{[nt]}}{A_n\sqrt{Q_n}}\right)^2\right] \leq C\eta.$$

Since $V_n/Q_n \rightarrow 1$ in probability, this gives tightness at the origin with the canonical denominator $A_n\sqrt{V_n}$. Thus the environment is admissible and tight at the origin. Theorem 1 yields the functional limit after canonical normalisation; multiplying by $A_n\sqrt{V_n}/\sqrt{n} \rightarrow (1-2a)^{-1/2}$ in probability gives the stated CLT and FCLT. □

5.2 Critical environments

Corollary 2 (Critical ERW classification). *For the predictable elephant random walk, assume that there is a deterministic constant $D_c < \infty$ such that*

$$\sum_{n=1}^{\infty} \frac{|a_n - 1/2|}{n} \leq D_c, \quad a_n \leq \frac{1}{2},$$

and $-1 + \varepsilon \leq a_n \leq 1 - \varepsilon$. Then

$$A_n = Ln^{1/2}(1 + o(1)) \quad \text{for some } L \in (0, \infty), \quad Q_n \sim L^{-2} \log n.$$

Consequently

$$\frac{S_n}{\sqrt{n \log n}} \Rightarrow \mathcal{N}(0, 1),$$

and the linearly interpolated process satisfies

$$\left(\frac{S_{[nt]}}{\sqrt{n \log n}} \right)_{0 \leq t \leq 1} \Rightarrow (\sqrt{t} N)_{0 \leq t \leq 1} \quad \text{in } C[0, 1],$$

where $N \sim \mathcal{N}(0, 1)$.

Proof. The product estimate is the logarithmic analogue of Lemma 4; the summability assumption makes $A_n/n^{1/2}$ converge to a finite positive limit. Hence

$$Q_n = \sum_{k < n} A_{k+1}^{-2} \sim L^{-2} \log n.$$

The second moment recursion gives $E[S_n^2] \leq Cn \log n$. Therefore

$$\mathbb{E} \left[\frac{R_n}{Q_n} \right] \leq \frac{C}{\log n} \sum_{k < n} \frac{\log k}{k^2} \rightarrow 0.$$

For every $\eta > 0$, uniformly for $t \in [\eta, 1]$,

$$\frac{A_{[nt]}}{A_n} \rightarrow \sqrt{t}, \quad \frac{Q_{[nt]}}{Q_n} \rightarrow 1.$$

By Lemma 2, $V_n/Q_n \rightarrow 1$ in probability and the same local bracket ratios hold for V . The jump condition follows from

$$\max_{k \leq n} \frac{|\Delta M_k|}{\sqrt{V_n}} \leq \frac{C}{\sqrt{\log n}} \rightarrow 0,$$

and the preceding estimate on R_n/Q_n also holds uniformly on $[\eta, 1]$. The origin condition is

$$\mathbb{E} \left[\sup_{0 \leq t \leq \eta} \left(\frac{S_{[nt]}}{\sqrt{n \log n}} \right)^2 \right] \leq C\eta \frac{\log(n\eta + 2)}{\log n}.$$

Since $A_n \sqrt{V_n} \sim \sqrt{n \log n}$ in probability, this is exactly the tightness at the origin required by Definition 1. The limiting process is $G(t) = \sqrt{t} B(1)$, with $G(0) = 0$, which is continuous. Theorem 1 completes the proof. $\square$

5.3 Superdiffusive environments

Corollary 3 (Superdiffusive ERW classification). *For the predictable elephant random walk, if $a_n \rightarrow a > 1/2$ and there is a deterministic constant $D_s < \infty$ such that*

$$\sum_{n=1}^{\infty} \frac{|a_n - a|}{n} \leq D_s,$$

then $A_n = Ln^a(1 + o(1))$ for some $L \in (0, \infty)$, $V_{\infty} < \infty$, and

$$\frac{S_n}{n^a} \longrightarrow LM_{\infty} \quad \text{almost surely and in } L^2.$$

Proof. The product estimate gives $A_n = Ln^a(1 + o(1))$. The deterministic bound $\sum |a_n - a|/n \leq D_s$, together with the bounded Taylor remainders in $\log(1 + a_n/n)$, also gives a deterministic lower bound

$$A_n \geq cn^a \quad (n \geq 1)$$

for some $c > 0$. Hence

$$Q_{\infty} \leq c^{-2} \sum_{n \geq 1} n^{-2a} < \infty$$

deterministically. Since $V_n \leq Q_n$, also $V_{\infty} < \infty$ and $\mathbb{E}[V_{\infty}] < \infty$. The finite-bracket part of Theorem 1 applies. $\square$

6 Why the Classification Is Optimal

Proposition 1 (Regularly varying ERW clock diagram). *For the predictable elephant random walk, suppose that the product of a bounded environment satisfies*

$$A_n \sim Cn^{\alpha}L(n),$$

where $C \in (0, \infty)$, L is slowly varying, and the self-interaction correction is negligible whenever $Q_n \rightarrow \infty$. Then the product clock $Q_n = \sum_{k < n} A_{k+1}^{-2}$, and hence the ERW bracket whenever $R_n = o(Q_n)$, has the following three regimes.

1. *If $\alpha < 1/2$, then*

$$Q_n \sim \frac{n}{(1 - 2\alpha)A_n^2}, \quad A_n \sqrt{Q_n} \sim \sqrt{\frac{n}{1 - 2\alpha}},$$

so the canonical Gaussian scale is diffusive.

2. *If $\alpha = 1/2$, then*

$$Q_n \sim C^{-2} \sum_{k < n} \frac{1}{kL(k)^2}.$$

The critical normalisation is therefore determined by the slowly varying integral, not by the exponent alone.

3. *If $\alpha > 1/2$, then $Q_{\infty} < \infty$, and the walk is in the finite-bracket martingale regime.*

Proof. This is Karamata's theorem applied to $A_{k+1}^{-2} \sim C^{-2}k^{-2\alpha}L(k)^{-2}$. The cases $\alpha < 1/2$, $\alpha = 1/2$, and $\alpha > 1/2$ correspond respectively to a divergent power sum, a borderline slowly varying sum, and a convergent sum. The classification theorem then translates these bracket alternatives into the corresponding limit regimes. $\square$

Proposition 2 (Coefficient convergence does not classify products). *Fix $a < 1/2$ and $c \in \mathbb{R}$. The deterministic environment*

$$a_n = a + \frac{c}{\log(n+2)}$$

can be chosen bounded away from -1 and 1 , satisfies $a_n \rightarrow a$, and

$$A_n \sim C_c n^a (\log n)^c \quad \text{for some } C_c \in (0, \infty).$$

Thus the limiting coefficient a does not determine the martingale normalisation.

Proof. The Taylor expansion of the logarithm gives

$$\log A_n = \sum_{k < n} \frac{a}{k} + c \sum_{k < n} \frac{1}{k \log(k+2)} + O(1) = a \log n + c \log \log n + O(1).$$

Exponentiating proves the claim. $\square$

Proposition 3 (Different invariants give different canonical laws). *Let $a < 1/2$. For the constant environment $a_n \equiv a$ and the slow environment $a_n = a + c/\log(n+2)$, $c \neq 0$, the terminal diffusive CLT has the same variance $(1-2a)^{-1}$, but the canonical martingale normalisations differ by the factor $(\log n)^c$. Hence no classification based only on the limiting coefficient can be pathwise complete.*

Proof. Both environments satisfy Lemma 4, so Corollary 1 gives the same terminal CLT. Proposition 2 shows that their products are not asymptotically equivalent unless $c = 0$. Since $S_n = A_n M_n$, the martingale-scale description retains this distinction even when the terminal $\sqrt{n}$ -CLT does not. $\square$

Proposition 4 (Logarithmic critical trichotomy). *Let*

$$a_n = \frac{1}{2} + \frac{c}{\log(n+2)}$$

for all large n . Then

$$A_n \sim C n^{1/2} (\log n)^c.$$

Consequently the bracket scale is

$$Q_n \asymp \begin{cases} (\log n)^{1-2c}, & c < 1/2, \\ \log \log n, & c = 1/2, \\ 1, & c > 1/2. \end{cases}$$

Consequently the canonical normalisation $A_n \sqrt{Q_n}$ has the following orders:

$$A_n \sqrt{Q_n} \asymp \begin{cases} \sqrt{n \log n}, & c < 1/2, \\ \sqrt{n \log n \log \log n}, & c = 1/2, \\ \sqrt{n} (\log n)^c, & c > 1/2. \end{cases}$$

Thus the boundary is not determined by the pointwise limit $a_n \rightarrow 1/2$, but by the product-bracket geometry; the same limiting coefficient produces diffusive logarithmic scaling, iterated-logarithmic critical scaling, or finite-bracket supercritical behaviour according to the value of c .

Proof. The product asymptotic is the calculation of Proposition 2 with $a = 1/2$. The displayed alternatives follow from the integral test applied to

$$Q_n \sim C^{-2} \sum_{k < n} \frac{1}{k (\log k)^{2c}}.$$

Multiplying these estimates by $A_n^2 \sim C^2 n (\log n)^{2c}$ gives the orders of $A_n \sqrt{Q_n}$. $\square$

Remark 3. These examples are deliberately deterministic. They show that the obstruction is not caused by random learning noise or measurability subtleties. It is already present in the product geometry of the simplest predictable environments.

7 Examples of Predictable Environments

Learning rules are not part of the main theory; they are sources of predictable environments. The classification theorem applies once the product and bracket geometry has been identified.

Proposition 5 (Lipschitz policy maps). *Let θ_n be $\mathcal{F}_n$ -measurable, let $\ell : \mathbb{R}^d \rightarrow (\varepsilon, 1 - \varepsilon)$ be Lipschitz, and put*

$$p_n = \ell(\theta_n), \quad a_n = 2p_n - 1.$$

If $\theta_n \rightarrow \theta_\star$ almost surely, $a = 2\ell(\theta_\star) - 1 < 1/2$, $a_n \leq \bar{a} < 1/2$, and the induced product is uniformly diffusive in the sense of Definition 2, then the diffusive classification applies. If there is a deterministic constant $D_\theta < \infty$ such that

$$\sum_{n=1}^{\infty} \frac{\|\theta_n - \theta_\star\|}{n} \leq D_\theta,$$

then the corresponding stable diffusive, critical or superdiffusive product asymptotics follow according to the value of a .

Proof. Lipschitz continuity gives

$$|a_n - a| \leq 2 \operatorname{Lip}(\ell) \|\theta_n - \theta_\star\|.$$

The claims are direct applications of the phase diagram. □

Remark 4 (On regret and stochastic approximation). Algorithms such as projected stochastic approximation, confidence-bound procedures, exponential weights or posterior sampling enter the present theory only after they have been proved to generate a predictable coefficient path with one of the product–bracket geometries above. The paper does not use regret as a black box and does not claim a universal theorem for these algorithms.

Remark 5 (Numerical diagnostics). The accompanying scripts reproduce finite-sample checks for the diffusive, critical, superdiffusive and slowly varying regimes. These computations are not used as evidence for the theorems; they are retained only to make the examples and normalisations reproducible.

8 Discussion and Open Problems

The classification suggests several problems beyond the scalar predictable environment treated here.

Multidimensional predictable environments. For walks on $\mathbb{Z}^d$ with increments taking values among the coordinate directions and their opposites, the same scalar product A_n normalises the conditional drift whenever $\mathbb{E}[X_{n+1} \mid \mathcal{F}_n] = a_n S_n / n$. The remaining issue is the limiting directional covariance of the martingale increments. A complete vector classification should combine the scalar product–bracket geometry with a matrix-valued covariance invariant.

Continuous-time analogues. The product A_n is the discrete solution of a logarithmic-time linear equation. Continuous-time memory processes should lead to multiplicative functionals of predictable drift environments and bracket clocks of the same kind.

Random memory kernels. The present paper keeps the recalled time uniform and puts all randomness in the repeat–reverse environment. If the recall kernel itself becomes predictable, the conditional drift is no longer $a_n S_n/n$. Identifying the right invariants for that problem would extend the classification beyond scalar products.

Quenched and annealed universality. The results here are pathwise in the realised predictable environment. A more ambitious theory would distinguish quenched limits conditional on an external environment from annealed limits when the environment has its own dynamics. Feedback environments, where a_n is generated from the walk itself, are the main obstruction to a direct random-media analogy.

Reinforced and self-interacting models. The product–bracket mechanism may also appear in reinforced walks or adaptive urns after linearisation around a stable direction. The open question is whether there is a broader martingale classification principle behind these models, or whether the complete-memory ERW is a special solvable case.

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